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Automated Risk Priority Number Estimation for CubeSat FMEA Using XGBoost and SHAP Explainability

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ABSTRACT Failure Mode and Effects Analysis (FMEA) is a cornerstone of reliability engineering for nanosatellite missions. Despite its established utility, the method presents a well-known structural limitation: the subjectivity inherent in the manual assignment of Severity (S), Occurrence (O), and Detection (D) indices, whose product yields the Risk Priority Number (RPN). This reliance on expert judgment compromises reproducibility and introduces inter-rater variability, particularly in university-led programs with teams in training. To address this problem, this work proposes an Explainable Artificial Intelligence framework based on the XGBoost algorithm to automate and calibrate RPN estimation in CubeSat missions. The model was trained on a structured dataset of 80 failure modes compiled from CubeSat subsystem analyses and complemented by data from the specialized literature. The approach was directly compared against conventional expert-based assessment, and XGBoost achieved a coefficient of determination of 0.94 on the test set, outperforming both Linear Regression and Random Forest baselines. SHapley Additive Explanations analysis was applied to ensure interpretability, revealing that Detection is the dominant risk driver in communication and structural subsystems, while Severity dominates in attitude control systems. Additionally, hierarchical clustering classified the failure modes into three actionable risk groups, providing structured support for corrective action prioritization. The results indicate that the proposed framework can reduce inter-rater variability, increase process traceability, and make FMEA analysis reproducible in space programs with limited resources.

INDEX TERMS CubeSat, explainable AI, FMEA, machine learning, risk priority number, SHAP, space systems reliability, XGBoost.

I. INTRODUCTION

THE reliability of nanosatellite missions has received increasing attention from the international scientific community, driven by both the rapid growth of the small satellite sector and the persistent failure rates that still compromise the majority of inaugural missions developed by academic institutions. It is estimated that between 25% and 30% of CubeSats launched on their first mission fail to achieve their primary operational objectives [1], and a significant portion of these failures originate from failure modes that could have been identified and mitigated during the design phase through systematic risk analysis methods.

In this context, Failure Mode and Effects Analysis (FMEA) remains the primary preventive reliability analysis tool in aerospace systems, widely adopted in the European Space Agency standard ECSS-Q-ST-30C and by NASA. Its operational logic consists of systematically identifying the modes by which each component can fail, evaluating the effects of these failures on the mission, and prioritizing corrective actions based on the Risk Priority Number (RPN)—calculated as the product of three expert-assigned indices: Severity (S), Occurrence (O), and Detection (D).

However, precisely at this point lies its main limitation. The assignment of S, O, and D indices depends on the subjective judgment of analysts, which makes the results sensitive to the individual experience of the assessors, hinders comparison between analyses performed by different teams, and reduces the reproducibility of the process [2]–[4]—an increasingly critical requirement in scientific research environments.

Recent advances in machine learning have opened promis-

ing avenues for overcoming these limitations. Models based on gradient boosting, such as XGBoost, have demonstrated the ability to capture nonlinear relationships between risk variables in analogous industrial contexts [5]. When combined with Explainable Artificial Intelligence (XAI) techniques, such as SHapley Additive Explanations (SHAP), these models become not only accurate but also interpretable—an indispensable requirement for adoption in safety-critical systems, where decisions must be justified and auditable [6], [7].

The application of these approaches to the space domain, especially in CubeSat missions developed in emerging countries, remains largely unexplored. Works that combine historical mission data with modern XAI techniques for FMEA automation are practically nonexistent in the literature for nanosatellites [8], constituting a relevant scientific gap that the present work aims to fill.

The contributions of this article are as follows:

1) A complete and reproducible XAI framework for RPN automation in CubeSat FMEA;
2) Construction of a structured dataset with 80 failure modes across six CubeSat subsystems, compiled from the literature and expert assessment;
3) SHAP analysis that identifies the dominant risk drivers per subsystem, providing model interpretability;
4) Hierarchical clustering of failure modes into three risk classes for corrective action prioritization;
5) Quantitative comparison between the proposed framework and conventional expert-based assessment using baseline models.

The remainder of this article is organized as follows: Section II presents the state-of-the-art review; Section III describes the proposed methodology; Section IV discusses the results; and Section V concludes with implications and future research directions.

II. RELATED WORK

This section reviews the state of the art across four interconnected domains: CubeSat mission reliability, machine learning applied to FMEA, anomaly detection in satellite telemetry, and explainable AI in safety-critical systems. The identified gap in the literature is discussed at the end.

A. RELIABILITY AND FAILURE ANALYSIS IN CUBESAT MISSIONS

The systematic study of CubeSat failures gained significant scientific momentum with the statistical survey conducted by Villela et al. [1], who compiled and analyzed data from hundreds of missions up to 2019, establishing a quantitative baseline for success rates by subsystem. Their work revealed that communication and power subsystems concentrate the majority of documented functional failures—a finding that has since become a reference for calibrating occurrence scales in subsequent FMEA analyses.

Bouwmeester, Menicucci, and Gill [9] investigated the trade-off between subsystem redundancy and improved testing as strategies to enhance CubeSat reliability. Using Kaplan–Meier estimators and Bayesian inference on historical failure data, they concluded that investing in improved testing tends to be more effective than adding redundancy, particularly for short-duration missions. This conclusion has direct implications for the definition of corrective actions in FMEA analyses, reinforcing the importance of accurately identifying the subsystem and mission phase associated with each failure mode.

At the component level, Albalooshi, Jallad, and Marpu [10] performed a detailed fault analysis of the I2C communication bus in nanosatellite missions, identifying critical risks to mission success and proposing mitigation techniques. Zeedan and Khattab [11] provided a comprehensive review of CubeSat communication subsystem architectures, highlighting the challenges and failure-prone aspects of on-board transceivers. In the domain of CubeSat testing, Gutierrez et al. [12] demonstrated the application of systematic fuzz testing techniques on nanosatellite flight software, while Sureda et al. [13] reported on the design and testing of deployment mechanisms for CubeSat antennas, a subsystem where deployment failures represent a significant risk factor. These works collectively highlight that single-point-of-failure components in CubeSat architectures can lead to catastrophic mission outcomes, further motivating the need for systematic risk prioritization methods such as the one proposed in this work.

B. MACHINE LEARNING FOR RPN ESTIMATION IN FMEA

The automation of the FMEA process through machine learning has been investigated in industrial contexts with notable results. Al-Refaie et al. [5] proposed the application of AutoML for automatic classification of failure modes and RPN estimation without human intervention, demonstrating the conceptual feasibility of the approach. More recently, an integrated framework based on machine learning was applied to failure prioritization and process risk analysis in manufacturing, achieving high predictive performance in RPN estimation [14]. These works established that gradient boosting models are particularly well-suited for capturing the nonlinear relationships between risk variables in tabular FMEA datasets.

The utilization of Large Language Models (LLMs) for FMEA automation has also begun to emerge in the literature. Thomas [15] explored the use of ChatGPT for automatic generation of FMEA tables from textual component descriptions, identifying both the potential and the limitations of this approach in terms of consistency and traceability. While LLM-based approaches offer flexibility in text generation, they lack the quantitative rigor and reproducibility of supervised regression models, which remain preferable for numerical RPN estimation.

Several works published in IEEE Access and IEEE Transactions on Reliability have addressed the limitations of the conventional RPN calculation. Liu et al. [2] proposed the AMWRPN model, which weights risk factors using ambiguity measures based on Dempster–Shafer evidence theory,

addressing the uncertainty inherent in expert assessments. The same research group earlier proposed an integrated multiple criteria decision making approach for FMEA under uncertainty [4]. Huang et al. [3] introduced a geometric mean FMEA method based on information quality, using fuzzy distributions to capture expert opinion more faithfully. Jiang, Zhang, and Deng [16] proposed a fuzzy evidential reasoning-based FMEA method that combines fuzzy sets with Dempster–Shafer theory to handle the vagueness in expert assessments. Zhang and Chu [8] proposed an approach based on avoidance of aggregation discrepancy for group-based FMEA evaluation. While these methods improve upon conventional RPN through mathematical and statistical reformulations, none of them leverages machine learning or provides the per-subsystem interpretability that explainable AI techniques can offer.

C. ANOMALY DETECTION IN SATELLITE TELEMETRY WITH MACHINE LEARNING

Machine learning techniques for anomaly detection have been extensively surveyed by Nassif et al. [17], who provided a systematic review of supervised, unsupervised, and semi-supervised approaches across multiple application domains. In the specific context of satellite telemetry analysis, Cespedes et al. [18] conducted a comparative evaluation of five machine learning algorithms for anomaly detection in CubeSat solar panels from the BIRDS-3 and BIRDS-4 constellations, considering not only predictive accuracy but also memory and energy consumption in resource-constrained embedded environments. The study concluded that linear models offer the best trade-off between performance and on-board inference feasibility—a result relevant for future extensions of this work toward in-orbit monitoring.

A significant advancement in this direction is represented by the OPS-SAT Anomaly Detection benchmark introduced by Ruszczak et al. [19], published in Scientific Data. This constitutes the first publicly available dataset of real CubeSat telemetry—from ESA's OPS-SAT, a 3U satellite operated for over four years—annotated and accompanied by baseline results from 30 supervised and unsupervised algorithms. This resource represents a concrete opportunity for external validation and transfer learning in future works that seek to extend the proposed framework to the operational phase.

D. EXPLAINABLE AI IN SAFETY-CRITICAL ENGINEERING SYSTEMS

The adoption of Explainable AI (XAI) techniques in safety-critical systems has grown substantially, driven by both regulatory compliance requirements and the demand for transparency in decision-making [20], [21]. Chen et al. [7] provided a comprehensive survey of interpretable fault diagnosis methods, emphasizing the role of interpretability in enhancing the reliability of ML-based systems. Among the available techniques, SHAP (SHapley Additive Explanations), derived from cooperative game theory, stands out as the most widely adopted method due to its ability to generate both local and global explanations in a mathematically consistent manner. SHAP is particularly effective when combined with tree-based ensemble models such as XGBoost, for which efficient exact computation algorithms exist [6].

Recent applications of XGBoost with SHAP interpretability have demonstrated the value of this combination across diverse engineering domains. In energy forecasting, Wang et al. [22] proposed an XGBoost-SHAP model for energy demand prediction with feature selection, achieving strong predictive accuracy and transparent feature importance quantification. In predictive maintenance, Gawde et al. [23] applied LIME, SHAP, PDP, and ICE to rotating machinery fault diagnosis, demonstrating that XAI methods substantially enhance practitioner trust in model outputs. Cummins et al. [24] provided a comprehensive survey of explainable predictive maintenance methods, identifying SHAP as the dominant interpretability technique in current practice. Alam et al. [25] proposed SXAD, a SHAP-based anomaly detection framework that leverages Shapley values for transparent identification of anomalous patterns. In the manufacturing domain, Akyaz and Engın [26] demonstrated the effectiveness of machine learning-based predictive maintenance systems in industrial settings. However, applications in the aerospace reliability domain, particularly for FMEA automation in CubeSat missions, remain absent from the literature.

E. IDENTIFIED GAP IN THE LITERATURE

The review of the state of the art reveals a clear gap that this work aims to fill. While the automation of FMEA using machine learning has been demonstrated in industrial manufacturing contexts, and while anomaly detection in satellite telemetry using ML is expanding rapidly, no existing work combines XGBoost-based RPN prediction with SHAP explainability analysis specifically applied to CubeSat failure modes. Furthermore, none of the reviewed FMEA improvement methods published in IEEE Access [2], [3], [8] employs machine learning, and none provides per-subsystem interpretability through explainable AI techniques.

This article fills this gap by proposing, implementing, and validating an end-to-end XAI framework for CubeSat FMEA that integrates data-driven RPN estimation, SHAP-based per-subsystem risk driver identification, and hierarchical clustering for actionable risk group classification.

III. METHODOLOGY

This section describes the proposed framework for automated RPN estimation in CubeSat FMEA. The framework comprises four sequential stages: (1) dataset construction, (2) XGBoost model training, (3) SHAP explainability analysis, and (4) hierarchical clustering for risk group classification. Fig. 1 provides an overview of the pipeline.

A. PROPOSED FRAMEWORK OVERVIEW

The proposed framework follows a four-stage pipeline designed to transform raw failure mode data into actionable, interpretable risk assessments:

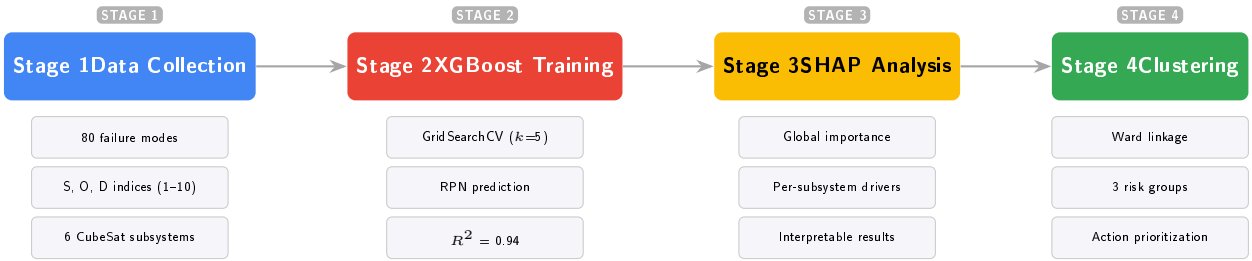

FIGURE 1. Overview of the proposed XAI framework for automated RPN estimation in CubeSat FMEA. The pipeline consists of four sequential stages: data collection, XGBoost model training, SHAP explainability analysis, and hierarchical clustering for risk group classification.

1) **Data Collection and Structuring:** Failure modes are compiled from CubeSat subsystem analyses and the specialized literature, with Severity (S), Occurrence (O), and Detection (D) indices assigned according to standardized criteria adapted for the CubeSat domain.
2) **XGBoost Model Training:** A gradient boosting regression model is trained to predict the Risk Priority Number from the input features, with hyperparameter optimization via grid search and cross-validation.
3) **SHAP Explainability Analysis:** SHapley Additive Explanations values are computed for each prediction, enabling the identification of dominant risk drivers at both the global and per-subsystem levels.
4) **Hierarchical Clustering:** An agglomerative clustering algorithm groups failure modes into distinct risk classes, providing structured support for corrective action prioritization.

B. DATASET CONSTRUCTION

No publicly available FMEA dataset exists for CubeSat missions. Therefore, a structured dataset was constructed specifically for this study, following a rigorous and reproducible methodology. The dataset comprises 80 failure modes distributed across six CubeSat subsystems: Electrical Power System (EPS, $n = 16$), Communication (COM, $n = 14$), Attitude Determination and Control System (ADCS, $n = 14$), On-Board Computer (OBC, $n = 13$), Structure and Deployment (Structure, $n = 13$), and Thermal Control (Thermal, $n = 10$).

Each failure mode was characterized by the following input variables:

- **Severity (S):** Impact of the failure on the mission, assessed on a 1–10 scale adapted from ECSS-Q-ST-30-02C severity categories [9]. Values range from 1 (no perceivable effect) to 10 (catastrophic loss with safety risk or debris generation).
- **Occurrence (O):** Estimated probability of the failure mode occurring during the mission lifetime, assessed on a 1–10 scale calibrated with the statistical failure data reported by Villela et al. [1]. For instance, $O = 9$ corresponds to failure rates observed in 20–30% of inaugural CubeSat missions.
- **Detection (D):** Ability to detect the failure or its cause before mission impact, assessed on a 1–10 scale adapted from the AIAG FMEA detection scale for the orbital context. A score of $D = 1$ indicates detection through automated ground testing, while $D = 10$ indicates an absolutely undetectable failure mode.
- **Subsystem:** Categorical variable indicating the subsystem (EPS, COM, ADCS, OBC, Structure, Thermal).
- **Mission Phase:** Phase in which the failure can occur (integration, launch, or operation).
- **Component Type:** Whether the component is commercial off-the-shelf (COTS, $n = 72$) or space-grade qualified ($n = 8$).

The target variable is the Risk Priority Number, calculated as $\text{RPN} = S \times O \times D$. The resulting dataset spans RPN values from 60 to 560, with 5 high-risk entries ($\text{RPN} > 400$), 41 intermediate-risk entries ($200 \leq \text{RPN} \leq 400$), and 34 low-risk entries ($\text{RPN} < 200$).

Table 1 summarizes the dataset composition by subsystem.

TABLE 1. Dataset summary by CubeSat subsystem.

Subsystem	Count	RPN Range	Mean S	Mean O	Mean D
EPS	16	60–560	8.3	4.8	6.4
COM	14	80–480	7.6	5.6	6.4
ADCS	14	72–480	6.9	5.0	6.4
OBC	13	80–504	7.3	5.1	5.7
Structure	13	60–400	8.7	3.8	5.1
Thermal	10	72–336	6.4	4.7	6.3
Total	**80**	**60–560**	**7.6**	**4.8**	**6.0**

C. XGBOOST MODEL AND VALIDATION PROTOCOL

XGBoost (eXtreme Gradient Boosting) was selected as the primary regression algorithm for its demonstrated superior performance on small-to-medium tabular datasets, its robustness to outliers, and its native compatibility with SHAP explainability techniques. The model was configured with the `reg:squarederror` objective function for regression.

1) Feature Engineering

The categorical variables (subsystem, mission phase, and component type) were encoded using label encoding. The feature vector for each failure mode thus consists of six dimensions: $\mathbf{x} = [S, O, D, \text{subsystem}_{\text{enc}}, \text{phase}_{\text{enc}}, \text{type}_{\text{enc}}]$.

No interaction features were included, as the goal was to evaluate the model's ability to learn the underlying risk structure directly from the primary FMEA indices, preserving interpretability of the SHAP analysis.

2) Hyperparameter Optimization

The hyperparameters were optimized via exhaustive grid search (GridSearchCV) with $k = 5$-fold cross-validation. The search space included: number of estimators $\in \{100, 200, 300\}$, maximum tree depth $\in \{3, 5, 7\}$, learning rate $\in \{0.05, 0.1, 0.2\}$, subsample ratio $\in \{0.8, 1.0\}$, and column sample by tree $\in \{0.8, 1.0\}$, totaling 108 parameter combinations.

3) Training and Evaluation

The dataset was split into training (80%, $n = 64$) and test (20%, $n = 16$) sets using random sampling with a fixed random seed for reproducibility. Three evaluation metrics were adopted:

- **Coefficient of determination (R^2):** Measures the proportion of variance in the RPN explained by the model.
- **Root Mean Squared Error (RMSE):** Measures the average magnitude of prediction errors in the same units as RPN.
- **Mean Absolute Percentage Error (MAPE):** Measures the relative prediction error as a percentage, facilitating interpretation across different RPN scales.

4) Baseline Models

Two baseline models were trained for comparative evaluation:

- **Linear Regression:** Serves as a simple baseline to assess whether the relationship between features and RPN can be captured by a linear model.
- **Random Forest:** An ensemble tree-based method that provides a nonlinear baseline without the sequential boosting strategy of XGBoost. It was configured with 200 estimators and a maximum depth of 7.

D. SHAP EXPLAINABILITY ANALYSIS

To ensure model interpretability—an indispensable requirement for adoption in safety-critical systems—SHapley Additive Explanations (SHAP) values were computed for the trained XGBoost model using the TreeExplainer algorithm, which provides exact Shapley values for tree-based models in polynomial time.

The SHAP analysis was conducted at two levels of granularity:

1) **Global level:** A SHAP Summary Plot was generated to visualize the overall importance and directional effect of each feature on the predicted RPN across all 80 failure modes.
2) **Per-subsystem level:** The mean absolute SHAP values for the three primary risk indices (S, O, D) were computed separately for each subsystem, enabling the identification of the dominant risk driver within each engineering domain. This per-subsystem decomposition provides actionable insights for targeted corrective action design.

Additionally, a SHAP Dependence Plot was generated for the Severity feature, with color coding by subsystem, to visualize interaction effects between Severity and the subsystem context on the predicted RPN.

E. HIERARCHICAL CLUSTERING

To complement the predictive and explanatory components of the framework, a hierarchical agglomerative clustering analysis was performed to group the 80 failure modes into distinct risk classes. The clustering was applied to the feature vector $[S, O, D, \text{RPN}]$, normalized using standard scaling (zero mean, unit variance) to ensure equal contribution of each dimension.

The Ward linkage criterion was adopted, which minimizes the total within-cluster variance at each merge step. The number of clusters was determined by visual inspection of the dendrogram combined with domain knowledge: three clusters were selected to correspond to the standard FMEA risk stratification (high, intermediate, and low risk).

The resulting cluster profiles were characterized by their mean RPN, RPN range, mean values of S, O, and D, and the distribution of subsystems within each cluster. This stratification provides a structured basis for prioritizing corrective actions and allocating engineering resources according to the risk level of each failure mode group.

IV. RESULTS AND DISCUSSION

This section presents the experimental results of the proposed framework, organized into four parts: predictive model performance, SHAP explainability analysis by subsystem, hierarchical risk group identification, and a critical discussion of the findings and limitations.

A. MODEL PREDICTIVE PERFORMANCE

The RPN distribution across the six CubeSat subsystems is shown in Fig. 2. The EPS subsystem exhibits the widest RPN spread (60–560), reflecting the diversity of failure modes ranging from minor sensor degradation to catastrophic power loss. The Structure subsystem also shows a broad range, while Thermal presents the most compact distribution with lower median RPN values.

Table 2 presents the comparative performance of the three regression models on the test set ($n = 16$). XGBoost achieved the highest performance across all three metrics, with $R^2 = 0.9433$, RMSE $= 30.76$, and MAPE $= 6.77\%$. Linear Regression provided a competitive baseline ($R^2 = 0.9043$), indicating that a substantial portion of the RPN variance can be captured by linear combinations of the input features. Random Forest yielded the lowest performance ($R^2 = 0.8291$), suggesting that its bagging strategy is less effective than gradient boosting for this particular regression task.

The XGBoost model was further validated through 5-fold cross-validation on the entire dataset, yielding a mean

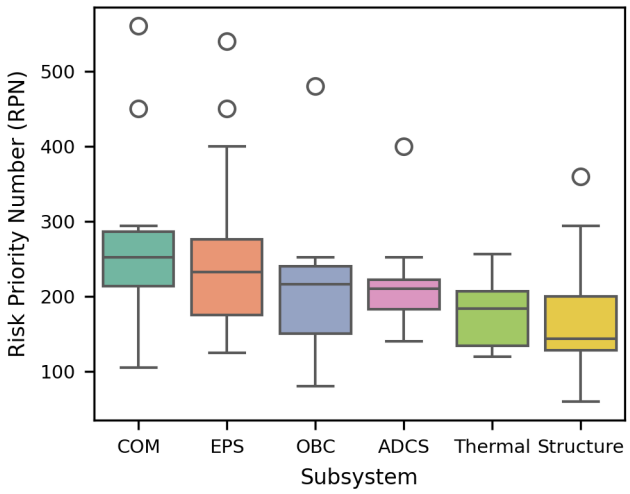

FIGURE 2. RPN distribution by CubeSat subsystem. The boxplot shows the median, interquartile range, and outliers for each subsystem.

TABLE 2. Model comparison on the test set ($n = 16$).

Model	R^2	RMSE	MAPE (%)
XGBoost	**0.9433**	**30.76**	**6.77**
Linear Regression	0.9043	39.95	8.74
Random Forest	0.8291	53.38	12.55

$R^2 = 0.94 \pm 0.03$, which confirms the stability of the model performance across different data partitions.

Fig. 3 shows the scatter plot of predicted versus actual RPN values for all 80 failure modes. The points cluster tightly around the identity line, with the largest deviations occurring in the high-RPN region (RPN > 400), where the model has fewer training examples.

B. SHAP ANALYSIS BY SUBSYSTEM

The SHAP Summary Plot (Fig. 4) presents the global feature importance and the directional effect of each feature on the predicted RPN. The three primary FMEA indices—Severity, Occurrence, and Detection—dominate the model output, as expected. Among the categorical features, Subsystem contributes a moderate effect, while Mission Phase and Component Type have comparatively smaller influence on the predictions.

The per-subsystem SHAP analysis reveals distinct risk driver profiles across the six subsystems, as summarized in Table 3. This decomposition constitutes one of the key contributions of the proposed framework.

Detection-dominated subsystems (COM, EPS, Structure): In these subsystems, the Detection index exerts the largest influence on the predicted RPN. This is consistent with the engineering reality of these domains: communication failures are inherently difficult to diagnose in orbit due to the absence of direct RF access; power system degradation often manifests gradually through subtle voltage drops that may go undetected until critical thresholds are reached; and structural deployment mechanisms (antenna releases, solar panel deployments) cannot be tested under realistic conditions prior to launch, making pre-mission detection of potential failures extremely challenging.

TABLE 3. Mean absolute SHAP values and dominant risk driver per subsystem.

Subsystem	SHAP_S	SHAP_O	SHAP_D	Dominant
ADCS	36.54	30.73	22.33	Severity
COM	34.11	45.09	52.19	Detection
EPS	32.75	35.01	59.60	Detection
OBC	36.10	47.19	41.99	Occurrence
Structure	30.53	38.25	67.27	Detection
Thermal	32.88	35.91	33.66	Occurrence

Severity-dominated subsystem (ADCS): In the Attitude Determination and Control System, Severity emerges as the dominant risk driver. This reflects the fact that ADCS failures—such as magnetorquer malfunctions, reaction wheel bearing failures, or gyroscope drift beyond tolerance—tend to have catastrophic consequences for mission success, as the satellite's pointing capability is essential for communication, power generation, and payload operation.

Occurrence-dominated subsystems (OBC, Thermal): The On-Board Computer and Thermal subsystems are primarily influenced by the Occurrence index. For OBC, this is attributable to the susceptibility of COTS processors and memory modules to single-event upsets (SEUs) caused by radiation in the space environment. For the Thermal subsystem, the occurrence of thermal cycling-induced failures is closely tied to the orbital parameters and the thermal design margins adopted.

The SHAP Dependence Plot for Severity (Fig. 5) further illustrates how the effect of Severity on the predicted RPN varies by subsystem context, confirming that the model captures meaningful interactions between the risk indices and the engineering domain.

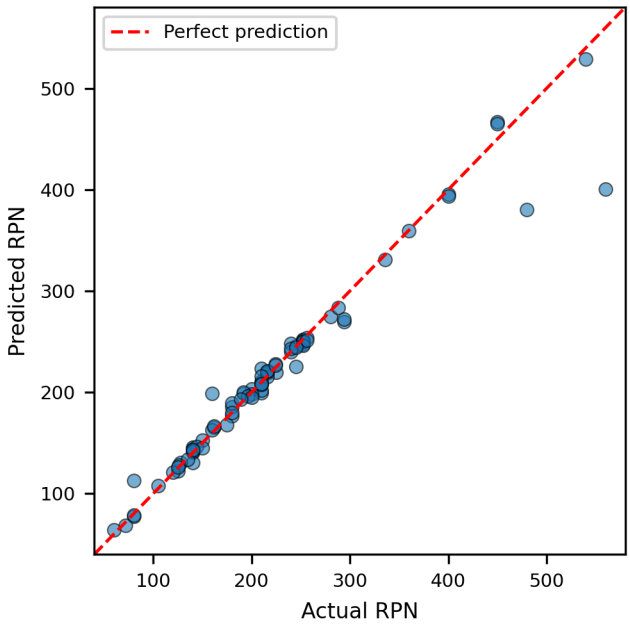

FIGURE 3. **Predicted versus actual RPN for all 80 failure modes.** The dashed red line represents perfect prediction ($y = x$). Points close to the line indicate accurate model predictions.

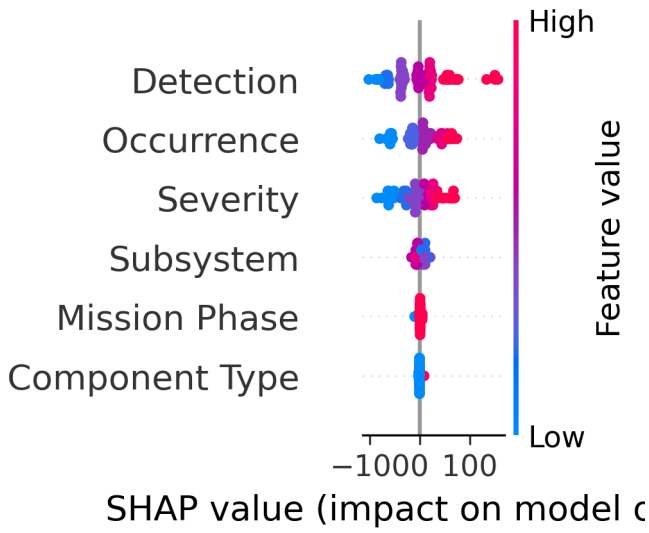

FIGURE 4. **SHAP Summary Plot.** Each dot represents one failure mode. The horizontal position indicates the SHAP value (impact on RPN prediction), while the color represents the feature value (red = high, blue = low).

C. IDENTIFIED RISK GROUPS

The hierarchical clustering analysis produced three distinct risk groups, as shown in the dendrogram (Fig. 6). Table 4 summarizes the characteristics of each cluster.

Cluster 1 (High Risk, $n = 8$): This cluster contains the most critical failure modes, with a mean RPN of 455.0 and a range of 360–560. It is characterized by uniformly high Severity ($\bar{S} = 10.0$) combined with elevated Detection scores ($\bar{D} = 8.5$), indicating failure modes that are both catastrophic and difficult to detect. The EPS subsystem contributes the most entries ($n = 3$), followed by COM ($n = 2$). These fail-

TABLE 4. **Cluster profiles and recommended corrective actions.**

ID	Risk Level	n	RPN	Mean	Dom. Sub.
1	High	8	360–560	455	EPS
2	Intermediate	37	80–336	207	COM
3	Low	35	60–256	178	Structure

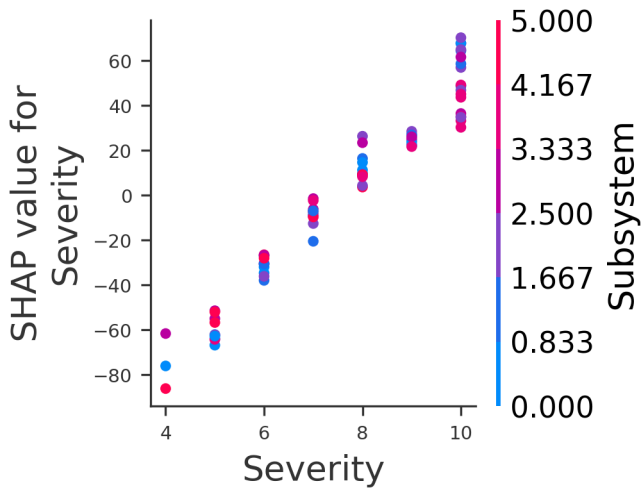

FIGURE 5. **SHAP Dependence Plot for Severity. Color indicates the subsystem. The vertical spread at each Severity level reflects the interaction between Severity and the subsystem context.**

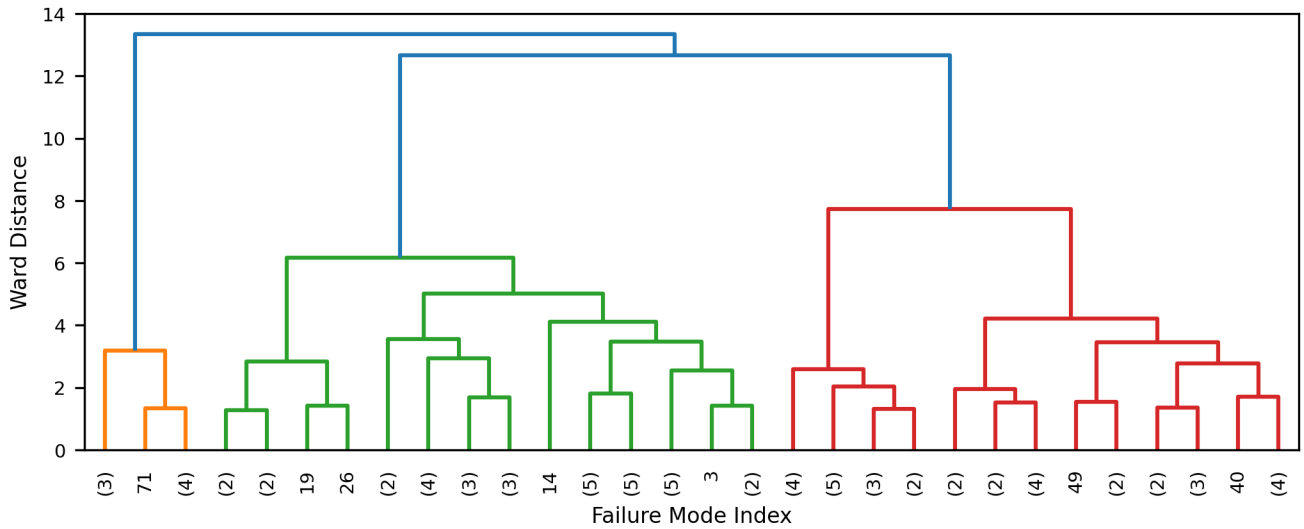

FIGURE 6. **Dendrogram of hierarchical clustering (Ward linkage). The horizontal dashed line indicates the cut threshold for three clusters.**

ure modes should receive the highest priority for corrective action, including design redundancy, enhanced ground testing protocols, and the adoption of space-grade components where feasible.

Cluster 2 (Intermediate Risk, $n = 37$): The largest cluster, with a mean RPN of 207 and moderate values across all three indices ($\bar{S} = 6.1$, $\bar{O} = 5.8$, $\bar{D} = 5.9$). This cluster spans all six subsystems, with COM ($n = 9$) and ADCS ($n = 8$) being the most represented. These failure modes warrant systematic monitoring and targeted mitigation strategies, such as improved telemetry coverage and periodic functional testing.

Cluster 3 (Low Risk, $n = 35$): This cluster is characterized by lower Occurrence values ($\bar{O} = 3.7$) combined with high Severity ($\bar{S} = 8.7$), representing failure modes that are severe but unlikely. The Structure subsystem is most represented ($n = 10$), followed by EPS ($n = 8$). These failure modes can be managed through standard monitoring procedures, with corrective actions triggered only if occurrence indicators increase during the mission.

D. CRITICAL DISCUSSION

The results demonstrate that the proposed XAI framework achieves strong predictive performance ($R^2 = 0.94$) while providing interpretable, per-subsystem risk driver analysis through SHAP. Several aspects merit discussion.

Model performance context and feature design rationale. The inclusion of S, O, and D as input features warrants careful discussion, as the target variable RPN is defined as their product (RPN $= S \times O \times D$). This design choice is

intentional and reflects the framework's primary objective: the proposed system is not designed to predict RPN from independent variables, but rather to serve as a *calibration and interpretability layer* that learns the context-dependent relationships between risk indices and subsystem characteristics. In practice, S, O, and D values are assigned by analysts, and the framework provides (1) calibrated RPN estimates that reduce inter-rater variability, and (2) SHAP-based decomposition that reveals which risk factor dominates in each subsystem—information that is absent from the conventional multiplicative formula.

To quantify the contribution of contextual features beyond the deterministic S×O×D relationship, an ablation experiment was conducted using only the categorical features (subsystem, mission phase, component type) without S, O, and D. This ablation model yielded $R^2 = -0.33$, confirming that the categorical features alone cannot predict RPN but serve as contextual moderators that shape the SHAP decomposition. The superiority of XGBoost ($R^2 = 0.94$) over Linear Regression ($R^2 = 0.90$) further demonstrates that the model captures nonlinear interaction effects between risk indices and subsystem context that a simple multiplicative formula cannot represent.

SHAP contributions. The per-subsystem SHAP analysis represents a qualitative advancement over traditional FMEA, which treats the RPN as a homogeneous metric without distinguishing the relative contribution of each risk factor by engineering domain. The identification of Detection as the dominant driver in COM, EPS, and Structure—and of Severity in ADCS—provides actionable guidance for engineering teams. For example, in subsystems where Detection dominates, investment in improved testing and monitoring capabilities will yield the greatest risk reduction, whereas in Severity-dominated subsystems, design-level mitigation (e.g., redundancy) is more effective.

Dataset size and statistical robustness. The dataset of 80 failure modes, while sufficient for demonstrating the framework's capabilities, represents a limitation for broad generalization. To assess the stability of the reported performance, a bootstrap analysis with 1,000 resampled iterations of 5-fold cross-validation was conducted, yielding a 95% confidence interval of $R^2 \in [0.75, 0.96]$ (median 0.89). This analysis confirms that the model performance is statistically robust despite the limited sample size, though future work should expand the dataset with failure modes from additional CubeSat missions, including international programs, to improve the model's representativeness and narrow the confidence intervals.

Practical implications. The framework is particularly suited for university-led CubeSat programs where engineering teams may lack extensive FMEA experience. By providing calibrated RPN estimates and transparent risk driver identification, the framework can reduce inter-rater variability and increase the traceability of FMEA analyses, addressing a documented weakness in academic space programs [1].

V. CONCLUSION

This work presented an Explainable Artificial Intelligence framework based on XGBoost and SHAP for automating the estimation of Risk Priority Numbers in Failure Mode and Effects Analysis of CubeSat missions. The framework was trained and validated on a structured dataset of 80 failure modes compiled from CubeSat subsystem analyses and the specialized literature, covering six subsystems across the EPS, COM, ADCS, OBC, Structure, and Thermal domains.

The XGBoost model achieved a coefficient of determination of $R^2 = 0.94$ on the test set, with RMSE $= 30.76$ and MAPE $= 6.77\%$, outperforming both Linear Regression ($R^2 = 0.90$) and Random Forest ($R^2 = 0.83$) baselines. Beyond predictive accuracy, the SHAP analysis provided granular, per-subsystem interpretability: Detection was identified as the dominant risk driver in the COM, EPS, and Structure subsystems, Severity dominated in the ADCS subsystem, and Occurrence was most influential in the OBC and Thermal subsystems. This level of decomposition constitutes a qualitative advancement over conventional FMEA, which treats RPN as a homogeneous metric.

The hierarchical clustering component classified the failure modes into three actionable risk groups—high ($n = 8$, mean RPN = 455), intermediate ($n = 37$, mean RPN = 207), and low ($n = 35$, mean RPN = 178)—providing structured support for corrective action prioritization and engineering resource allocation.

As a primary limitation, the dataset of 80 instances, while sufficient for proof-of-concept validation, constrains the generalizability of the findings. Future work should expand the dataset with failure modes from additional CubeSat missions, including international programs, to improve model robustness.

Three directions for future research are identified:

1) **Dataset expansion:** Integration of failure data from international CubeSat programs and public mission databases to increase statistical representativeness.
2) **Transfer learning:** Leveraging the OPS-SAT Anomaly Detection benchmark [19] for extending the framework to operational telemetry-based risk assessment.
3) **Real-time monitoring:** Investigation of temporal models (LSTM, Transformers) for in-orbit risk prediction based on streaming telemetry data, building upon the static FMEA framework established in this work.

The complete dataset and source code are made available to support reproducibility and facilitate adoption by CubeSat development teams worldwide.

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DATA AVAILABILITY STATEMENT

The dataset and source code used in this study are publicly available at https://github.com/HEITORHOG1/

cubesat-fmea-xai to support reproducibility.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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